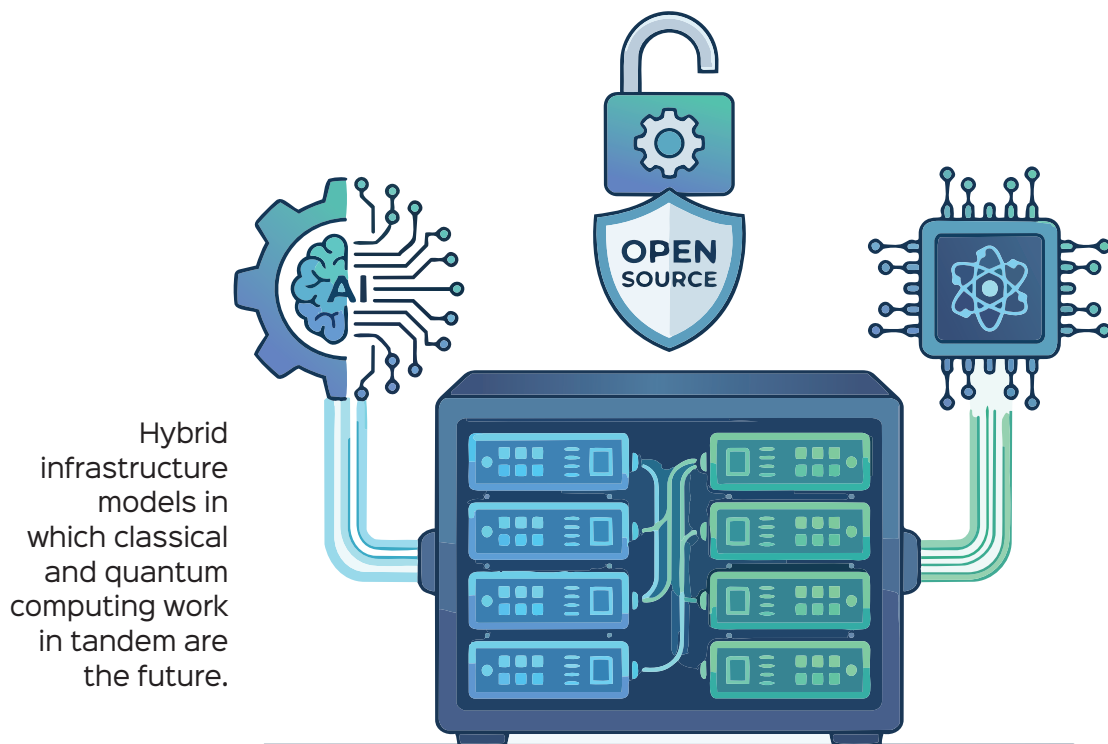


Building Hybrid AI-Quantum Infrastructure with Open Source



Over the past decade, cloud computing has revolutionised the way application development and deployment are carried out, and it is now possible to deploy infrastructure in just minutes. However, two technology revolutions are changing the landscape of computing and the way infrastructure needs to be built and deployed. These are: large language models (LLMs) and quantum computing.

The adoption of AI systems, especially LLMs, has led to an enormous need for computing infrastructure. Traditional infrastructure built on the concept of virtualization is no longer sufficient for AI workloads. These now need specialised AI-native clouds that can manage the complex

pipelines of computing and scale dynamically. However, simultaneously, a new technological frontier is unfolding. Quantum computing, which was previously accessible only through experimental environments, is slowly and steadily being made accessible through the cloud. Instead of deploying and running quantum processors locally, which is almost impossible due to the extreme operating environments necessary for running them, it is now possible to access them remotely through APIs running in the cloud. This is called Quantum Computing as a Service (QCaaS).

The union of AI infrastructures and quantum computing is leading to a new paradigm of hybrid computing

environments, where classical computing resources like CPUs and GPUs are being combined with new quantum processors. The role of open source technologies is significant in the unfolding of a new era of hybrid infrastructures. Kubernetes is being accepted as the *de facto* backbone of cloud-native infrastructures, and it is enabling organisations to orchestrate complex distributed environments. At the same time, open source technologies are playing a significant role in the democratisation of AI and quantum infrastructures. As organisations are increasingly exploring the union of AI and quantum technologies, a new paradigm of infrastructure is unfolding, where AI clouds, HPC environments,